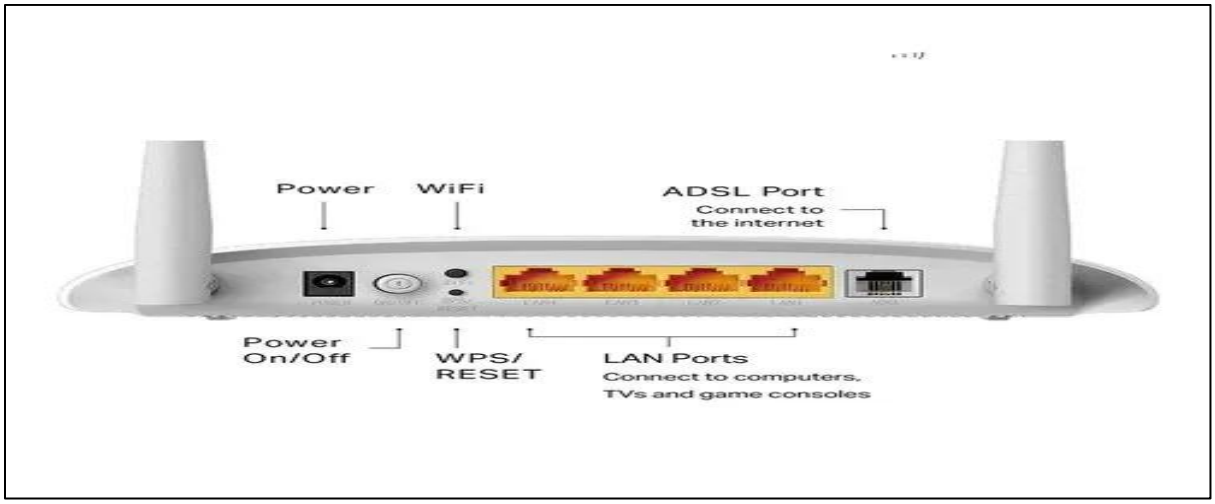
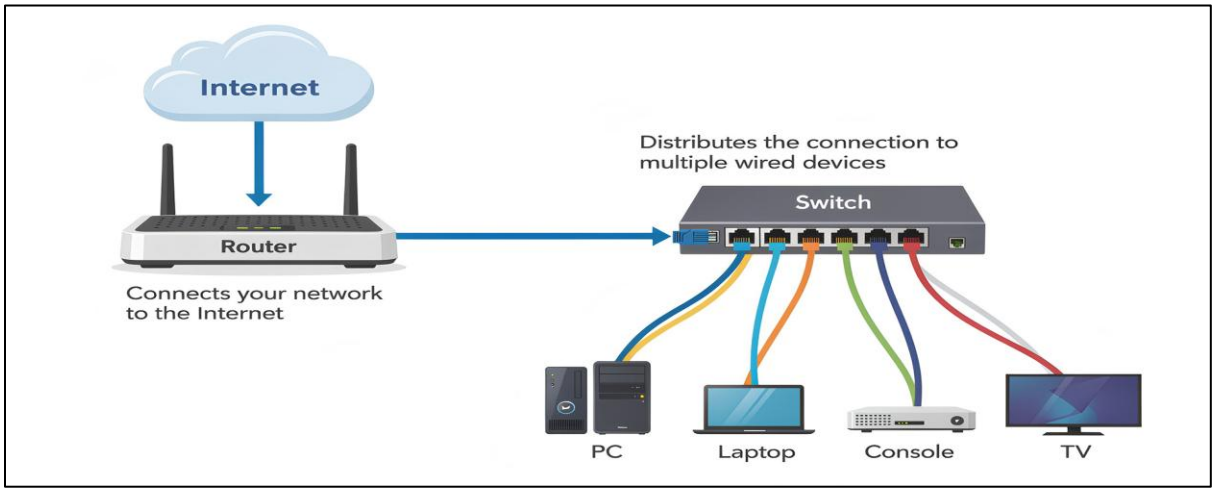
Assignment

Module 3.2 Router and switches Theory

1. **Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.**

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2. **List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.**

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IP Addresses vs. MAC Addresses: A router sends data between different networks using IP addresses, while a switch sends data between devices on the same local area network (LAN) using MAC addresses.

Main Function: A router links to different networks and sends data packets where they need to go. A switch links to devices within a LAN and sends data to the right device.
Handling Network Traffic: A router moves traffic between networks, like between a home network and the internet. A switch sends frames to certain devices on the same network by using its MAC address table.

3. **Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.**

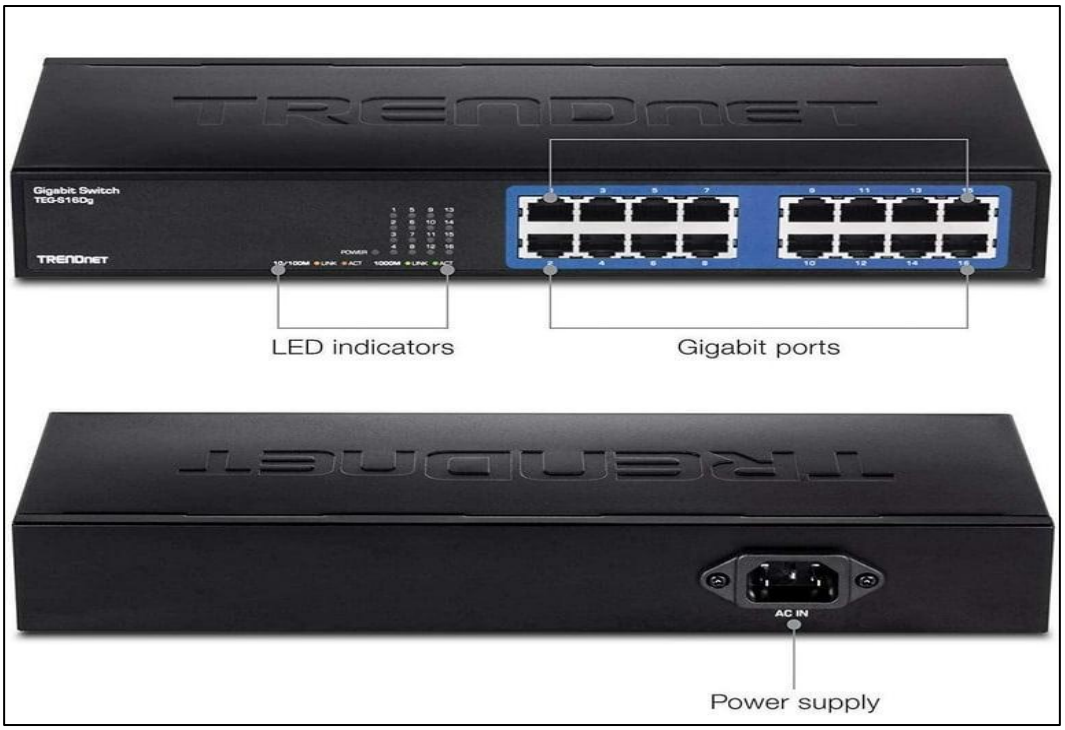
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Power Port: Generally found at the back of the router, it connects to a power adapter to supply electricity to the device.

Ethernet Ports: Usually positioned on the rear panel, the WAN port connects the router to the modem or internet source, while the LAN ports connect wired devices such as computers.

LED Indicators: Commonly located on the front or top of the router, these lights display the power status, internet connection, Wi-Fi activity, and Ethernet connectivity.



Power Port: Supplies electricity to the network switch so it can function properly.

Ethernet Ports: Enable computers and other devices to communicate through the same network.

LED Indicators: Show whether the switch is powered on and indicate connection and data activity.

4. **Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.**

- I would initially establish a connection between the router and the ISP line, as this component is responsible for delivering internet connectivity to the café's network infrastructure. Subsequently, I would link the router to an 8-port or 16-port switch by means of an Ethernet cable.

Following this step, I would connect each of the eight gaming personal computers to the switch using individual Ethernet cables. This configuration enables computers to establish communication with one another and obtain internet access.

Network layout:

ISP Line → Router → Switch → 8 Gaming PCs

The router serves as the gateway connecting the café to the internet, whereas the switch facilitates connectivity among all eight computers within the local network. The implementation of wired Ethernet connections ensures a more dependable and expeditious connection characterized by reduced latency, thereby rendering it appropriate for IPL streaming and multiplayer gaming applications.